

Lab 40 — Capstone: Complete Machine Learning Pipeline with Visualisation

Aim

To integrate NumPy, Pandas, machine learning models, evaluation, and visualisation into a complete end-to-end machine learning pipeline using the Iris dataset.

Algorithm / Procedure

1. Load the Iris dataset.
2. Convert the dataset into a Pandas DataFrame.
3. Display the dataset summary.
4. Split the data into training and testing sets.
5. Train a Decision Tree and KNN classifier.
6. Calculate the accuracy of both models.
7. Compare their performance.
8. Plot a bar chart showing model accuracies.
9. Save the plot as model_comparison.png.

Program

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score

iris=load_iris()

df=pd.DataFrame(iris.data,columns=iris.feature_names)
```

```

df['target']=iris.target

print("Dataset summary:\n",df.describe())

X_train,X_test,y_train,y_test=train_test_split(df.iloc[:,:-
1],df['target'],test_size=0.3,random_state=1)

models={'Decision Tree':DecisionTreeClassifier(random_state=1),'KNN
(k=5)':KNeighborsClassifier(n_neighbors=5)}

results={}

for name,model in models.items():

    model.fit(X_train,y_train)

    results[name]=accuracy_score(y_test,model.predict(X_test))

print("Model comparison:",results)

plt.bar(results.keys(),results.values())

plt.ylabel('Accuracy')

plt.title('Model Comparison - Iris Dataset')

plt.savefig('model_comparison.png')

plt.show()

```

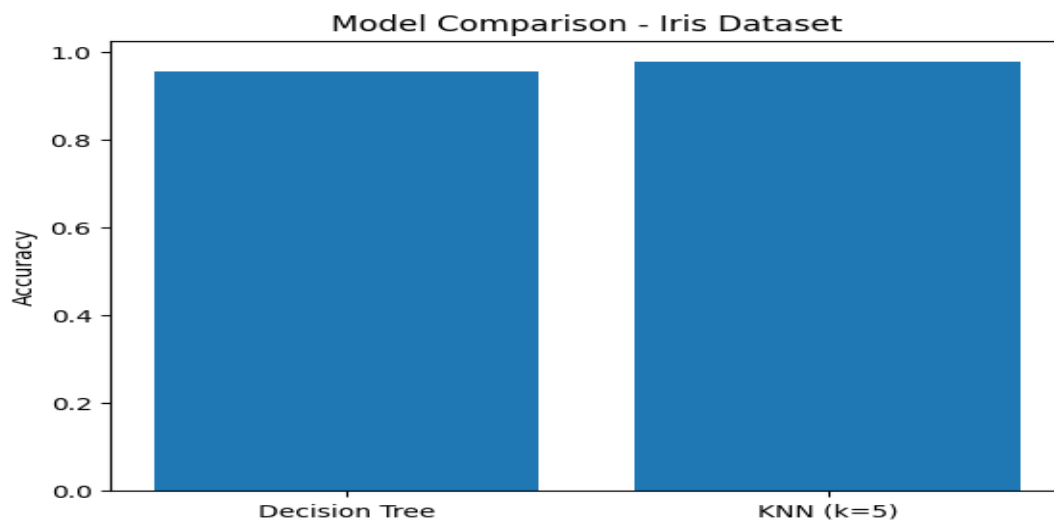
Output

Dataset summary:

	sepal length (cm)	sepal width (cm)	petal length (cm) \
count	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000
std	0.828066	0.435866	1.765298
min	4.300000	2.000000	1.000000
25%	5.100000	2.800000	1.600000
50%	5.800000	3.000000	4.350000

75%	6.400000	3.300000	5.100000
max	7.900000	4.400000	6.900000
petal width (cm) target			
count	150.000000	150.000000	
mean	1.199333	1.000000	
std	0.762238	0.819232	
min	0.100000	0.000000	
25%	0.300000	0.000000	
50%	1.300000	1.000000	
75%	1.800000	2.000000	
max	2.500000	2.000000	

Model comparison: {'Decision Tree': 0.9555555555555556, 'KNN (k=5)': 0.9777777777777777}



Result

A complete machine learning pipeline involving data exploration, model training, evaluation, comparison, and visualisation was successfully implemented. KNN achieved higher accuracy (97.8%) than the Decision Tree (95.6%) on the Iris dataset.